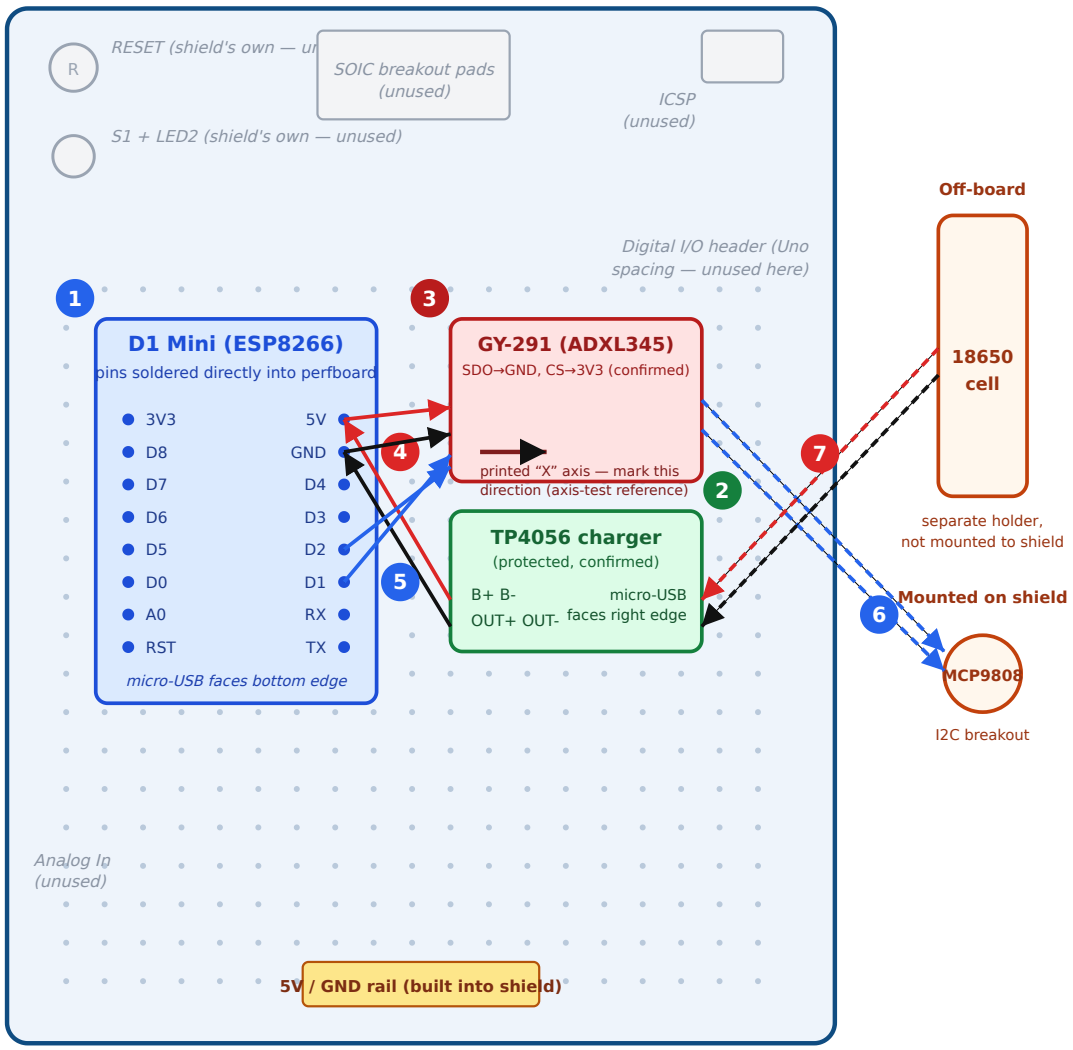
ESP8266 Bench Rig — Prototype Shield Build Guide

Temp + Gravity module bench prototype: D1 Mini (ESP8266) + TP4056 charger + GY-291 (ADXL345) + MCP9808 (I2C temperature) mounted on an Arduino-style Prototype Shield v.5.

This shield does not plug into the D1 Mini. It's Arduino Uno-shaped with Uno-spaced headers — a completely different pin pitch from the D1 Mini's own compact layout. Its Digital I/O / Analog In / ICSP headers go unused here. What's actually useful is its perfboard prototyping area, which you populate and wire by hand.



Prototype Shield v.5 (perfboard side up)

Legend

- Power (3V3 / 5V / BAT+)
- Ground
- Signal (I2C)

Before you start — already confirmed on this hardware

- ✅ TP4056 breakout is **protected** (3 ICs: TP4056 + DW01A + FS8205A)
- ✅ GY-291 `SDO` / `CS` are dedicated pins, not internally tied
- ✅ D0-to-RST "SLEEP" jumper pad is **soldered** — recommend a quick multimeter continuity check before your first real deep-sleep cycle
- ✅ A0 has an onboard divider (~220kΩ/100kΩ) — the external divider below (10kΩ/22kΩ) is sized for it
- ✅ Temperature sensor is an MCP9808 I2C breakout, not a DS18B20 probe — shares the GY-291's I2C bus, no extra GPIO needed

Step-by-step assembly

1. **Mount the D1 Mini.** Push its two 8-pin header rows straight down into two adjacent rows of the shield's perfboard grid (pin pitch matches exactly). Solder each pin from underneath. Orient it so the micro-USB port faces an open edge of the shield — you'll want it accessible for reflashing.
2. **Mount the TP4056 charger breakout.** Position it near a board edge so its micro-USB charging port stays reachable from outside the finished assembly. Tack it down (hot glue or double-sided tape is fine — it doesn't need to be soldered flat, just held still).
3. **Mount the GY-291 and mark its axis.** Position it away from the charger section (keeps switching noise off the I2C lines). **Before securing it, note which way the board's printed "X" axis arrow points** — mark this on the shield itself (a sharpie arrow works). This is your reference for the firmware's axis-calibration step later; repositioning the board after calibration means redoing that step.
4. **Wire the charger to the D1 Mini.** Two short wires: TP4056 `OUT+` → D1 Mini `5V`, and TP4056 `OUT-` → D1 Mini `GND`. The D1 Mini's onboard regulator runs fine from the cell voltage directly — no separate LDO needed.
5. **Wire the D1 Mini to the GY-291.** Four wires: D1 Mini `3V3` → GY-291 `VCC`, `GND` → `GND`, `D2` → `SDA`, `D1` → `SCL`. **Also wire GY-291 `SDO` → `GND` and `CS` → `3V3`** (both are dedicated pins on this board, not internally tied — confirmed from your photo).
6. **Wire the MCP9808 breakout.** Four wires, same pattern as the GY-291: MCP9808 `VDD` → D1 Mini `3V3`, `GND` → `GND`, `SDA` → the same `SDA` wire as the GY-291 (or directly to D1 Mini `D2`), `SCL` → the same `SCL` wire as the GY-291 (or directly to `D1`). It shares the GY-291's I2C bus — no new GPIO needed. Leave `ALERT` (if present) unconnected.
7. **Wire the battery.** 18650 cell (in its own holder, off the shield) → TP4056 `B+` / `B-`. Keep the holder separate rather than gluing it down — you'll be swapping/recharging it often during calibration.

Steps 4-7 can be done in any order — the numbering matches the diagram's callouts for cross-reference, not a strict sequence. Just avoid powering anything on until every wire in a given connection is in place (e.g. don't connect the battery with only one of B+/B- wired).

Power-on and first boot

8. **First power-up: USB only, no battery yet.** Plug the D1 Mini's own micro-USB into your computer. Open a serial monitor at 115200 baud. You should see boot messages — if nothing prints, check the USB cable isn't charge-only and that you selected the right serial port.

9. **Flash the firmware.** Open `firmware/fermentation_tilt_v1_esp8266/fermentation_tilt_v1_esp8266.ino`, fill in the CONFIG section (WiFi, Supabase URL/key, device ID/secret), select board **"LOLIN(WEMOS) D1 R2 & mini"** in the Arduino IDE, and upload.
10. **Run the axis test.** Hold the assembled rig upright (GY-291's marked X-axis vertical) — the serial log should print a tilt near 0°. Lay it flat — near 90°. If it's inverted or pinned near 90° in both positions, the firmware's axis assumption doesn't match your mounting; swap which axis `readTiltDegrees()` treats as the long one (comment in the code marks the line). Do this before any calibration data collection — a wrong axis makes every later reading garbage.
11. **Connect the battery, confirm deep sleep actually wakes.** Wire the cell in (step 7), let the unit go through one full sleep/wake cycle (default 900s), and confirm it reconnects and posts a reading. This is the real test of the D0-RST jumper — if the board never comes back, that joint needs reflowing.

Reference tables

Connection	From	To	Notes
Power	TP4056 OUT+	D1 Mini 5V	Same node as BAT+
Ground	TP4056 OUT-	D1 Mini GND	
Power	D1 Mini 3V3	GY-291 VCC	
I2C	D1 Mini D2	GY-291 SDA	GPIO4
I2C	D1 Mini D1	GY-291 SCL	GPIO5
Address set	GY-291 SDO	GND	I2C address 0x53
Mode set	GY-291 CS	3V3	Forces I2C mode
Power	D1 Mini 3V3	MCP9808 VDD	Shares GY-291's 3V3 wire
Ground	D1 Mini GND	MCP9808 GND	Shares GY-291's GND wire
I2C	D1 Mini D2	MCP9808 SDA	Same net as GY-291 SDA
I2C	D1 Mini D1	MCP9808 SCL	Same net as GY-291 SCL
ADC	BAT+	D1 Mini A0	via 10kΩ/22kΩ external divider
Wake	D0	RST	"SLEEP" pad, already soldered

Part	Qty	Notes
10kΩ resistor	1	Battery divider top (R1)
22kΩ resistor	1	Battery divider bottom (R2)
18650 cell + holder	1	~2500mAh
Jumper wires	~15	Solid-core preferred for perfboard soldering

See `SPEC.md`, `architecture/net_plan.md`, and `architecture/design_risks.md` for the full electrical derivation and design history behind these connections.